Summary in Graph

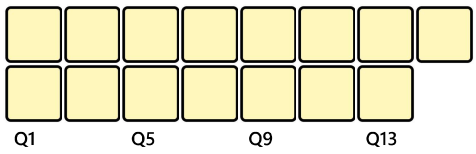
Exam Summary (GO Classes DA Test Series 2026 | Linear Algebra | Topic Wise Test 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (48) out of 479 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

If A is a 10×3 matrix that has a singular value decomposition (SVD) $U\Sigma V^T$, with

$$\Sigma = \begin{pmatrix} 100 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ then what is the rank of matrix } A \text{ ?}$$

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 06sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Suppose A (3 by 4) has the Singular Value Decomposition

$$A = U\Sigma V^{\mathrm{T}} = [\,u_1 \quad u_2 \quad u_3\,] \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} [\,v_1 \quad v_2 \quad v_3 \quad v_4\,]^{\mathrm{T}}.$$

What is the value of $\|Av_1\|$?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be an $m \times n$ real matrix, and let $A = U\Sigma V^T$ be any Singular Value Decomposition (SVD) for A, with U and V representing orthogonal matrices of sizes $m \times m$ and $n \times n$ respectively.
Let
$U = [\,\mathbf{u}_1 \quad \cdots \quad \mathbf{u}_r \quad \cdots \quad \mathbf{u}_m\,]$
$V = [\,\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_r \quad \cdots \quad \mathbf{v}_n\,]$
Let $r = \mathrm{rank}(A)$.
Which of the following assertions concerning the fundamental subspaces of A is accurate?

A. $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ forms an orthonormal basis for the column space of A.
B. $\{\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m\}$ forms an orthonormal basis for the null space of A.
C. $\{\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n\}$ forms an orthonormal basis for the null space of A^T.
D. $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ forms an orthonormal basis for the row space of A.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the 2×2 matrix

$$A = \frac{1}{\sqrt{10}} \begin{pmatrix} 2 \\ 1 \end{pmatrix} (\,1 \quad -1\,) + \frac{2}{\sqrt{10}} \begin{pmatrix} -1 \\ 2 \end{pmatrix} (\,1 \quad 1\,).$$

Which of the following statements is true about the singular values of matrix A ?

A. A has singular values $\sigma_1 = 2$ and $\sigma_2 = 1$.
B. A has singular values $\sigma_1 = \frac{1}{\sqrt{10}}$ and $\sigma_2 = \frac{2}{\sqrt{10}}$.
C. A has singular values $\sigma_1 = \frac{1}{\sqrt{10}}$ and $\sigma_2 = 1$.
D. A has singular values $\sigma_1 = \frac{2}{\sqrt{5}}$ and $\sigma_2 = \frac{1}{\sqrt{5}}$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square, invertible matrix and the Singular Value Decomposition (SVD) of $\mathbf{A}$ be

$$\mathbf{A} = \sum_{i=1}^{n} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

Which of the following steps are correct statements? (Select all that apply.)
In all options, $\mathbf{U} = [\mathbf{u}_1\mathbf{u}_2 \dots \mathbf{u}_n]$ and $\mathbf{V} = [\mathbf{v}_1\mathbf{v}_2 \dots \mathbf{v}_n]$ are orthogonal matrices, and $\mathbf{\Sigma}$ is a diagonal matrix with singular values σ_i on the diagonal.

A. The SVD of $\mathbf{A}$ can be written in matrix form as $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$.
B. The inverse of $\mathbf{A}$, denoted as $\mathbf{A}^{-1}$, can be found by taking the inverse of each component in the SVD: $\mathbf{A}^{-1} = \left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\right)^{-1} = \mathbf{V}\left(\mathbf{\Sigma}^{-1}\right)\mathbf{U}^T$.
C. The expression for $\mathbf{A}^{-1}$ can be rewritten as $\mathbf{A}^{-1} = \sum_{i=1}^{n} \frac{1}{\sigma_i}\mathbf{u}_i\mathbf{v}_i^T$, using the outer product of the columns of $\mathbf{V}$ and $\mathbf{U}$.
D. The expression for $\mathbf{A}^{-1}$ can be rewritten as $\mathbf{A}^{-1} = \sum_{i=1}^{n} \frac{1}{\sigma_i}\mathbf{v}_i\mathbf{u}_i^T$, using the outer product of the columns of $\mathbf{V}$ and $\mathbf{U}$.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Given an orthonormal basis $q_1, \dots, q_4$ for $\mathbf{R}^4$ and an orthonormal basis $z_1, \dots, z_6$ for $\mathbf{R}^6$, you create the 6×4 matrix $A = z_1q_1^T + z_2q_2^T$.
Which of the following is/are solutions of $Ax = z_1$? (Select all that apply.)

A. $x = q_1$
B. $x = q_1 + q_3$
C. $x = q_1 + q_4$
D. $x = q_1 + q_3 + q_4$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

The matrix $A = \begin{pmatrix} 1 & 0 \\ C & 1 \end{pmatrix}$.
Suppose σ_1 and σ_2 are the two singular values of A. What is $\sigma_1^2 + \sigma_2^2$?

A. $1 + C^2$
B. $2 + C^2$
C. $1 + 2C^2$
D. C^2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

For each of the following, determine whether a real 2×2 matrix A exists that satisfies the given conditions. Select all options that are possible:

A. There exists a matrix A such that A has determinant -1 and singular values 1 and 1 .
B. There exists a matrix A such that A has eigenvalues 1 and 1 and singular values 1 and 0 .
C. There exists a matrix A such that A has eigenvalues 0 and 0 and singular values 0 and 1 .
D. There exists a matrix A such that A has rank $r = 1$ and determinant 1 .

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider H, a symmetric matrix satisfying $H^2 = 4I$, where I denotes the 4x4 identity matrix. What is the product of all singular values of H ?

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let A be a 3×3 real matrix which transforms the following unit vectors:

$$x = \frac{1}{\sqrt{2}}(1,1,0) \mapsto Ax = (2,0,0)$$
$$x = \frac{1}{\sqrt{2}}(-1,1,0) \mapsto Ax = (0,-5,0)$$
$$x = (0,0,1) \mapsto Ax = (0,0,1)$$

Which of the following are the singular values of A ?

A. 2
B. 5
C. 1
D. -2

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

If A is a 3×3 non-singular real matrix with singular values $\sigma_1, \sigma_2, \sigma_3$, then which of the following options correctly provides the formulas for $|\det(A)|$?

A. $|\det(A)| = \sigma_1\sigma_2\sigma_3$
B. $|\det(A)| = \sigma_1^2\sigma_2^2\sigma_3^2$
C. $|\det(A)| = \sqrt{\sigma_1\sigma_2\sigma_3}$
D. $|\det(A)| = (\sigma_1\sigma_2\sigma_3)^{1/3}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

An invertible $\boldsymbol{n} \times \boldsymbol{n}$ matrix $\boldsymbol{A}$ has $\boldsymbol{n}$ distinct non-zero singular values. How many singular value decompositions $A = U\Sigma V^\top$ does A have?

A. 2^{n-1}
B. n^2
C. $n!$
D. 2^n

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider a matrix A, which is just a column vector, and its Singular Value Decomposition (SVD) as follows:

$$A = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & \vdots & 0 \\ 0 & \vdots & 1 \end{pmatrix} \begin{pmatrix} 1 \\ \cdots \\ 0 \end{pmatrix} (1)$$

What will be the projection matrix on the row space of A ?

A. $P_{\mathcal{R}(A)} = (1)$
B. $P_{\mathcal{R}(A)} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$
C. $P_{\mathcal{R}(A)} = (0)$
D. $P_{\mathcal{R}(A)} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the following statements regarding singular value decomposition (SVD) and related matrix properties. Which of the following is/are true?

A. Let $A = U\Sigma V^T$ be the singular value decomposition (SVD) of a square matrix A. Then $A+I = U(\Sigma + I)V^T$ necessarily represents the SVD of $A + I$, where I is the identity matrix of appropriate dimensions.
B. If A is an invertible matrix and σ is a singular value of A, then $\frac{1}{\sigma}$ is a singular value of A^{-1}.
C. For an $m \times n$ matrix A, the number of pivot columns in the row echelon form (REF) of A is equal to the number of nonzero singular values of A.
D. Suppose Z is an orthogonal matrix and $A = ZB$, where A and B are real matrices. Then, A and B have identical singular values because pre-multiplication by an orthogonal matrix preserves the singular values.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Select the statement(s) that are true regarding the properties of matrices and their singular values and eigenvalues:

A. The left singular vectors of A are eigenvectors of A^TA and the right singular vectors are eigenvectors of AA^T.
B. For any matrix A, the matrices AA^T and A^TA have the same eigenvalues.
C. If S is symmetric, then the nonzero eigenvalues of S are its singular values.
D. If A does not have full column rank, then 0 is eigenvalue of A. (Consider A as a square matrix)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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